Scheduling Learned Image Codes over Fading Multi-User Downlinks: A Link Abstraction and a Calibration-Aware Restless-Bandit Scheduler

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***Abstract*—Deep joint source-channel coding (JSCC) is designed and evaluated on single links, yet a cellular downlink serves many users slot by slot, through a scheduler, before a deadline. Scheduling needs each slot's value to each user; no model provides it for a learned code. We show that a progressive learned code trained under slot-varying SNR obeys a link abstraction: its distortion is the source distortion plus a sensitivity-weighted sum of per-slot minimum mean-square-error terms. On two test sets it predicts held-out decodes within 0.055–0.083 dB, against 0.16–0.19 dB for EESM and 0.29–0.34 dB for MIESM given the same weights; a constant-SNR-trained code obeys no such rule. Calibrated from 12 decodes per image, the abstraction reduces each user to a two-number state, and deadline scheduling becomes a finite-horizon restless bandit. We propose CLASP, a calibration-aware, link-abstraction-based scheduler that solves it with exact per-user dynamic programs, a Lagrangian bound on every causal scheduler and an index policy, and rewards the probability of reaching a quality target under the measured error of its own prediction, from the abstraction and reports. Decoded, CLASP gains 0.5–0.6 dB of mean PSNR over proportional fair and brings an extra 8–15% of users to 30 dB over per-image rate selection, with fresh, noisy or aged reports. Against digital transmission with CQI-table link adaptation and HARQ it leads by 1.2–1.6 dB and 13–16% of users at 30 dB under 2 dB CQI errors or aged reports. Training the code on the channel CLASP creates adds 0.21 dB.**

***Index Terms*—Deep joint source-channel coding, hybrid automatic repeat request, image transmission, imperfect channel state information, index policies, Lagrangian relaxation, link abstraction, multi-user scheduling, restless bandits.**

I. Introduction

DEEP joint source-channel coding (JSCC) maps an image straight to channel symbols with one learned encoder and decoder, with no separate source and channel codes [1]. It matches or beats separation-based digital transmission at low signal-to-noise ratio (SNR) and degrades gracefully as the channel worsens, instead of failing once the SNR drops below what a digital code was designed for [1], [2]. That makes it a strong candidate for image delivery in 6G [2]. But a learned code is designed and evaluated as a single link at a given SNR: one transmitter, one receiver, the channel condition handed to it [1], [3], [4]. A cellular downlink is different. A base station splits a frame of slots among many users through a scheduler; each user's slots fade independently; the base station knows only an estimate of the current slot, a channel-quality report [5] that is often late or wrong; and each delivery, one image, has a deadline. So the gain a point-to-point study reports is not automatically the gain a cell delivers. That depends on the scheduler, and no existing scheduler knows what a slot is worth to a learned code. To schedule, the base station needs the value of "give user k this slot now." For a digital code that value follows from the rate the slot carries and the codec's rate-distortion curve. For a trained, nonlinear code no such value exists.

Existing approaches to serving images to many users over a fading downlink fall into two categories (Fig. 1): *physical link design*, which keeps the image codec separate and engineers the link that carries its bits, and *the JSCC way*, which lets a learned code take on the channel itself.

Physical link design is how cellular systems deliver images today: a standard codec compresses the image into bits once, and the physical layer carries them with link adaptation from the user's channel quality indicator (CQI) reports, hybrid automatic repeat request (HARQ), and a channel-aware scheduler such as proportional fair (PF) [6]–[8]. Its tools for many users are mature (Section II-A), but they rest on a binary outcome per block, decoded or lost, so a user's quality is a step function of its channel: a slot below the block's design SNR delivers nothing, and SNR above it is wasted, the cliff effect [1], [2]. Moreover, the rate is committed before the slot is seen, from a report that may be late or wrong, and link adaptation and HARQ protect the bits only by spending extra slots. In our simulations, going from fresh reports to aged ones (correlation 0.8) costs the digital chain 1.9 dB of mean peak signal-to-noise ratio (PSNR), against 0.6 dB for a learned code under our scheduler (Section V-D).

In contrast, the JSCC way replaces the two codes with one learned code that maps the image straight to channel symbols [1]. Its quality degrades smoothly with the SNR its symbols meet, so no block is lost outright and no coding rate has to be matched to an uncertain report; the same symbols serve receivers of different SNRs, as linear analog codes showed for multicast video [9]. Most of this work adapts a single link to the channel and the budget (Section II-B). To serve many users, the JSCC way splits the shared channel by each user's predicted quality at its SNR [10]–[12], or learns the allocation by deep reinforcement learning [13], [14]. Our method belongs to the JSCC way: it schedules a learned code to many users across fading slots.

Although existing JSCC methods are effective, they face three challenges when a base station schedules a learned code to many users over fading slots.

- *No link abstraction.* They value a user's transmission at a

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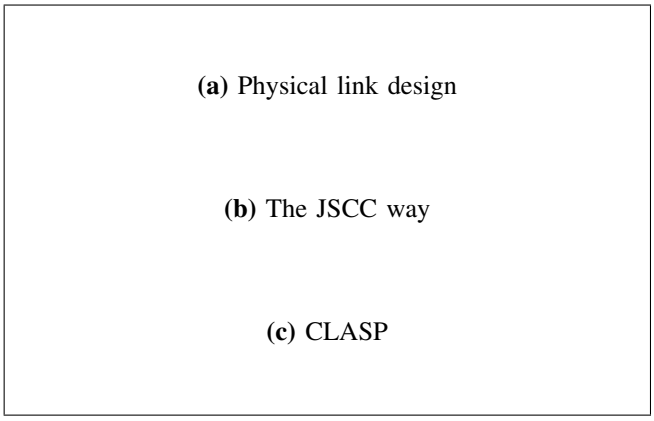


Fig. 1. Approaches to serving images to many users over a fading downlink. (a) Physical link design: a separate codec over adaptive modulation and coding, HARQ and a channel-aware scheduler; each block is decoded or lost. (b) The JSCC way: a learned code degrades gracefully, but each user is valued at a single SNR and allocated once, or scheduled by a policy learned by trial and error. (c) CLASP: a progressive learned code trained under slot-varying SNR obeys a link abstraction that values every slot, and a restless-bandit scheduler built on it serves every user slot by slot up to the deadline.

single SNR, with a fitted quality-SNR curve or a learned quality predictor [10], [12], [15]. A scheduled image, however, is spread over slots of different SNRs, and for a nonlinear learned code no rule is known for how these SNRs combine into decoded quality. The exponential and the mutual-information effective SNR mappings (EESM, MIESM) of physical link design predict a digital block's error rate from the SNRs of the resources it occupies [16], [17], not a learned code's distortion, and to our knowledge have not been validated on a learned code.

- *No tractable, certifiable scheduler.* Deciding which user each slot serves, before a deadline and without knowing future slots, is a stochastic dynamic program whose state grows exponentially with the number of users. Existing methods either decide each transmission at its current SNR without planning over the slots left before the deadline [10]–[12], or learn the allocation by reinforcement learning, which needs many interactions to train and gives no certificate of how far the learned policy is from optimal [13], [14].
- *Imperfect information.* The quality model and the channel reports are both estimates, yet existing schedulers take them at face value: per-image rate selection (PADC [15]) treats its predicted quality as exact, so a user predicted just above a quality target can decode just below it. Moreover, the advantage of the JSCC way over physical link design was measured on single links against an idealized digital side, with exact channel knowledge and capacity-achieving codes [1], [18]; whether it survives the reports a real base station has is open.

Behind all three lies one question: does a learned code have a link abstraction at all, a rule that, like EESM for a digital block, turns the SNRs its symbols met into its decoded quality? And if it does, what decides whether a code obeys it?

To answer it, we trained a progressive code, which decodes from any prefix of its phases, under slot-varying SNR, and tested candidate rules on held-out decodes with mixed slot SNRs (Sections IV-B and V-A). We found three things. First, its distortion adds up over slots: the source distortion plus a sensitivity-weighted sum of per-slot minimum mean-square-error (MMSE) terms predicts held-out decodes with a mean absolute error of 0.055–0.083 dB on two test sets, against 0.16–0.19 dB for EESM and 0.29–0.34 dB for MIESM with the same per-phase weights. Linear analog codes obey this form by construction [9]; here a nonlinear learned code acquires it. Second, training decides it: the same architecture trained at one constant SNR per image obeys no such rule (the MMSE rule errs by 0.22 dB, and no rule does better than 0.15 dB), because it codes its later phases against the base reconstruction it expects at one SNR, the mismatch DeepJSCC-l observed for layers [19]. Third, the abstraction is cheap and accurate, yet its small error matters. Twelve decodes per image calibrate it, and it predicts the decoded quality of complete scheduled deliveries with a mean absolute error of 0.06–0.09 dB and a bias within ± 0.03 dB, with exact, noisy or aged reports. At a hard quality target, however, a scheduler that takes the calibrated curves as exact leaves an extra 1.7–2.0% of the users below 30 dB, at every cell size from 16 to 128 users. In short, training gives a learned code a link abstraction. With it, each user's state reduces to two numbers, the phases received and their accumulated weighted noise, and scheduling becomes a planning problem, provided the abstraction's residual error is part of the objective.

To address these challenges, we propose CLASP (Calibration-aware, Link-Abstraction-based Scheduling of Progressive codes), a scheduler for learned progressive codes in a multi-user fading downlink. Unlike existing JSCC methods, which value each user at one SNR or learn the allocation by reinforcement learning, CLASP values every slot for every user through the code's link abstraction and plans each user's delivery to the deadline exactly. It has one design for each challenge.

- *Link abstraction.* The progressive code is trained under slot-varying SNR and its MMSE link abstraction is calibrated per image, so that each user becomes a two-number state.
- *Scheduler.* On this state the frame becomes a finite-horizon restless bandit [20]. A Lagrangian relaxation splits it into one small dynamic program per user under per-slot prices, solved exactly on a CPU; the relaxation's value is an upper bound on every causal scheduler; and users are served in the order of the Lagrangian index those programs define [21], [22].
- *Imperfect information.* Each program rewards the probability of reaching the quality target under a calibrated predictive distribution of the user's decoded quality, rather than a hard threshold on the predicted quality. The distribution's spread is the measured error of the prediction, from the calibrated curves and from the channel reports. Given an aged report, each program decides on the posterior of the channel, not on the report itself.

The main contributions of this paper are as follows.

- *Discovery.* We find that a learned progressive code trained

under slot-varying SNR obeys an additive MMSE link abstraction, more accurate than EESM and MIESM, while a code trained at constant SNR obeys none, and that a transmitter can calibrate it from 12 decodes per image. To our knowledge, effective-SNR mappings had not been tested on a learned code before.

- *Method.* We propose CLASP, which casts multi-user deadline scheduling of learned codes as a finite-horizon restless bandit with a two-number user state, solves it with exact per-user programs, a Lagrangian bound and an index policy, and is robust to the errors of its own quality prediction and of the channel reports.
- *Performance.* Decoded by the codec, CLASP gains 0.5–0.6 dB of mean PSNR over PF and brings an extra 8–15% of the users to 30 dB over per-image rate selection, with fresh, noisy or aged reports. In simulation its gap to the bound vanishes as the cell grows: from 0.21 to 0.005 dB per user between 8 and 128 users for mean PSNR, and for the target within $\pm 0.3\%$ of the bound from 16 users on. Against a digital chain with CQI-table link adaptation and HARQ, it leads at every CSI condition tested, by 1.2–1.6 dB and an extra 13–16% of the users at 30 dB with 2 dB CQI errors or reports aged to correlation 0.8. An idealized digital code 3 dB from capacity with the same HARQ leads CLASP by 2.4% of the users with exact reports and trails it by 5–14% from 1 dB CQI errors or correlation 0.95 on. Training the code on the channel CLASP creates adds 0.21 dB, decoded.

The rest of the paper is organized as follows. Section II reviews related work, and Section III states the system model and the problem. Section IV presents the training of the progressive code, its MMSE link abstraction, CLASP, and the training of the code on the channel CLASP creates. Section V evaluates the abstraction, and CLASP against PF, per-image rate selection, the Lagrangian bound and the digital chain. Section VI concludes and states the limitations.

II. Related Work

We review the two categories of Section I in turn, and then the scheduling theory CLASP builds on.

A. Physical Link Design for Multi-User Media Delivery

In a cellular downlink, an image is compressed once by a standard codec such as High Efficiency Video Coding (HEVC) [23], and its bits are carried by a physical layer that adapts to the channel. Adaptive modulation selects the constellation and the transmit power from the channel state, and its bit error rate depends on the error and the delay of the channel estimate it adapts to [6]; in Long Term Evolution (LTE), the receiver reports the channel as a 4-bit CQI [5]. HARQ retransmits a failed block, and the receiver either combines the noisy copies [24] or accumulates incremental redundancy; the throughput of such protocols over a fading Gaussian collision channel has closed forms [7]. Across users, a scheduler exploits multiuser diversity by serving each user when its instantaneous channel quality is near its peak, as PF scheduling does [8].

To evaluate such systems without simulating the decoding of every block, system-level simulators predict a block's error rate from the signal-to-interference-plus-noise ratios (SINRs) of all the resources it occupies. An effective SINR mapping compresses them into one effective SNR, from which the block error rate (BLER) is read off the link-level curve of an additive white Gaussian noise channel. EESM was derived for orthogonal frequency-division multiplexing (OFDM) from the union–Chernoff bound, with one parameter tuned per modulation and coding scheme [16]; MIESM maps the SINRs through the mutual information per received symbol or per bit [17]; and the 5G New Radio abstraction of the ns-3 simulator calibrates EESM and obtains SINR–BLER tables with a link-level simulator [25]. Both mappings predict whether a block decodes; Section V-A tests them as predictors of a learned code's distortion.

Media-aware schedulers value data by the distortion it removes. RaDiO reduces the rate-distortion optimized streaming of a presentation to the error-cost optimized transmission of each data unit, solved as a Markov decision process over the unit's transmission opportunities with a Lagrange multiplier pricing the rate, and essentially achieves the operational distortion-rate function of the source at the capacity of the packet erasure channel [26]. In a wireless downlink with several users, gradient-based schedulers set each user's rate from its channel quality and the gradient of a utility built from the distortion of its received video [27]. For packets that expire at the end of each frame, Hou, Borkar and Kumar characterize the feasible per-client delivery ratios and show that serving the client with the largest delivery debt first fulfills every feasible set [28]. All of these treat a block or a packet as decoded or lost, so the channel affects quality only through how many of them get through, the source of the cliff effect that learned codes avoid [1]. Our link abstraction takes the place of an effective SNR mapping for a learned code: it predicts distortion rather than a probability of decoding, and so it can value every slot a user receives.

B. The JSCC Way

Deep JSCC trains a convolutional encoder and decoder through a non-trainable channel layer. It outperforms JPEG or JPEG2000 compression with a capacity-achieving channel code at low SNR and small bandwidth, has no cliff effect, and degrades gracefully when the SNR departs from its training value [1]. Later codes adapt one model to the channel and the budget. Channel-wise attention scales the features by the SNR, so that one model serves a range of SNRs [3]; nonlinear transform source-channel coding learns an entropy model of its latent representation and adapts the transmission rate to it [29]; Swin Transformer backbones adapt to the channel state and the target rate within a single model [4]; and predictive and adaptive deep coding (PADC) predicts each image's PSNR from its content, the SNR and the compression ratio, and sends it at the least compression ratio that meets a quality constraint [15]. Other codes exploit the structure of the channel: OFDM over multipath fading [30], multiple-input multiple-output (MIMO) channels, with robustness to channel

estimation errors [31], and channel output feedback, which refines the receiver's estimate block by block and shortens variable-length transmission [18], [32]. DeepJSCC-l sends an image in layers, as successive refinements or as multiple descriptions, with negligible loss against a single transmission [19]. A tutorial on semantic and task-oriented communications is given in [2].

These codes are compared with separation-based transmission on single links: against JPEG or JPEG2000 with a capacity-achieving code [1], against Better Portable Graphics (BPG) with low-density parity-check (LDPC) codes over OFDM, with 2.5–4 dB of SNR gain at equal quality [30], and against BPG with 5G LDPC codes [4]. In some of these comparisons the digital side is idealized, with capacity-achieving codes and exact channel knowledge at the receiver [1], [18]; Section V-D compares instead with CQI-driven link adaptation and HARQ under erroneous and aged reports.

Linear analog transmission is the precedent for our link abstraction. SoftCast makes the network stack act as a linear transform of the pixels, so that channel noise reduces sharpness instead of causing glitches, and one multicast stream gives each receiver a quality commensurate with its channel [9]. For a linear code, the distortion is a sum of per-component MMSE terms by construction; the learned code of Section IV-B acquires this form, but only when it is trained for slot-varying SNR.

To serve many users, semantic and JSCC systems allocate the shared channel by each user's predicted quality: channel assignment and the number of transmitted semantic symbols from a semantic spectral efficiency [10]; user association and bandwidth from a message-level throughput [11]; compression ratios and transmission rates by Bayesian optimization over Gaussian-process models of quality against SNR and compression ratio [12]; and, in a non-orthogonal uplink shared with a bit-based user, the communication mode of the semantic user in each fading state [33]. Multi-user semantic transceivers are also trained end to end for image retrieval, machine translation and visual question answering [34]. Where the allocation changes over time, it is learned by deep reinforcement learning: proximal policy optimization with an attention network assigns resource blocks and the part of each user's semantic content to send [13], and a deep deterministic policy gradient agent sets each user's compression ratio, transmit power and bandwidth [14]. The allocations value each user's transmission at one SNR and decide it as a whole, so none of them can value a slot for an image whose symbols are spread over slots of different SNRs; the learned policies need many interactions to train and come with no bound on their distance from the optimum.

C. Restless Bandits and Lagrangian Index Policies

Whittle relaxed the constraint on the number of projects operated at each instant to a constraint on its expected value. The Lagrange multiplier of the relaxed constraint defines an index, and Whittle conjectured that operating the projects of largest index is nearly optimal when there are many of them [20]. Weber and Weiss proved the conjecture when the fluid approximation of the index policy has a globally stable equilibrium point, and gave an example in which it fails [35]. More generally, dynamic programs whose subproblems are coupled only through constraints on their actions admit Lagrangian and linear-programming relaxations whose value functions separate by subproblem [36]. Relaxing the limit on the number of items selected at a time yields both an upper bound on the optimal value and the Lagrangian index policy, and under mild conditions both are asymptotically optimal with many items, with tie-breaking playing an essential role [22]. A linear-program-based index policy is asymptotically optimal over finite and infinite horizons, at a square-root rate in general and exponentially fast under further conditions [37]. For deadline scheduling, Hao, Xu and Tong define the Lagrangian priority value of a job as the greatest tax under which it is optimal to process it, and prove that serving jobs in that order is asymptotically optimal as the system scales [21].

The same tools are established in wireless scheduling. For dynamic multichannel access, the Whittle index has a closed form, and for non-identical channels a Lagrangian relaxation gives an upper bound on performance [38]. For the age of information of clients behind unreliable channels, Whittle's index policy performs near optimally [39]. For remote inference, the joint choice of feature lengths and transmissions forms a restless multi-armed, multi-action bandit [40].

CLASP imports this machinery: the Lagrangian relaxation with one price per slot, its upper bound, and the Lagrangian priority policy [21], [22]. What is new is the arm and the reward: each user's phases accumulate over slots of different SNRs toward a decoded quality at the deadline, the link abstraction reduces each user to a two-number state whose program is solved exactly, and the reward accounts for the measured error of the quality prediction.

III. Preliminaries: System Model and Problem Statement

IV. Method

A. Training the Progressive Code for Slot-Varying SNR

B. The MMSE Link Abstraction

C. CLASP: Deadline Scheduling as a Finite-Horizon Restless Bandit

D. Imperfect Information: The Calibration-Aware Reward and Aged Reports

E. Training the Code on the Channel CLASP Creates

V. Experiments

A. Link Abstraction Accuracy

B. Mean PSNR: Near the Bound, and over PF

C. Quality Target and Latency

D. Imperfect CSI and the Digital Baseline

E. Training the Code on CLASP's Channel

F. Ablations and System Choice

VI. Conclusion

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